

Development of a vibrotactile stimulation system for cognitive rehabilitation

Master Thesis

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Abstract

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Keywords: Keyword 1, Keyword 2, Keyword 3, Keyword 4, Keyword 5

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1 Introduction

1.1 Motivation and Problem Statement

Dementia is one of the greatest medical and social challenges of the 21st century. More than 55 million people are affected worldwide - and the trend is rising [1], [2]. Despite advances in medication, treatment remains limited, particularly in terms of cognitive abilities.

Non-pharmacological interventions are therefore increasingly coming into focus [3]. Of particular interest are sensory-based approaches such as vibrotactile stimulation, which can activate specific areas of the brain via targeted stimuli. Initial animal studies show positive effects of 40 Hz stimulation on the reduction of amyloid deposits and on cognitive performance [4], [5], [6].

1.2 Objectives of the Thesis

The aim of this work is to develop a technical system for implementing such stimulation methods. The focus is on a precise, reproducible actuator system that is suitable for applications in the field of cognitive rehabilitation.

1.3 Structure of the Thesis

This work is theoretically and technically oriented and therefore begins with a chapter on the necessary basics. The aim is to provide the medical and technological background knowledge required to understand the system developed.

2 Theoretical Background

This chapter provides the theoretical foundation for the development of the vibrotactile stimulation system. It is divided into two main parts. The first part introduces the **medical background**, including neurodegenerative diseases such as Alzheimer's and Parkinson's disease, their associated cognitive impairments, and current approaches to cognitive rehabilitation. The second part focuses on the **technical background**, describing the principles of vibrotactile stimulation, actuation technologies, and existing systems. This structure ensures that both the clinical motivation and the technical requirements are clearly outlined before presenting the system analysis and implementation in the following chapters.

2.1 Medical Background

2.1.1 Neurodegenerative Diseases with Cognitive Impairments

Neurodegenerative diseases such as Alzheimer's disease and Parkinson's disease are among the leading causes of cognitive impairment in older adults. They are characterized by a progressive loss of neuronal structures and functions, which over time leads to marked deficits in memory, attention, executive abilities, and the capacity to manage everyday life [7].

Alzheimer's Disease

Alzheimer's disease is the most common form of dementia, accounting for approximately 60 % to 70 % of all cases [2]. According to the Alzheimer's Association, more than 55 million people worldwide are currently affected [1]. The disease is characterized by the pathological accumulation of proteins, including amyloid- β plaques and τ -protein-related neurofibrillary tangles, which lead to the progressive death of neurons [8].

The first noticeable symptoms usually involve a decline in episodic memory, the part of long-term memory responsible for personal experiences. As the disease advances, deficits in language, spatial orientation, and judgment become more pronounced. Alzheimer's disease is long-lasting and progressive, eventually resulting in complete dependence on care in the later stages [2].

2. Theoretical Background

Parkinson's Disease

After discussing Alzheimer's disease, the second major neurodegenerative disorder with cognitive impact is Parkinson's disease. Parkinson's disease is primarily known as a movement disorder, but cognitive symptoms are also frequent. Approximately 30 % to 40 % of patients develop Parkinson's dementia during the course of the disease [9]. Typical impairments include deficits in attention control, executive functions, and visuospatial perception, while memory is often less affected in the early stages [10].

The underlying cause of the disease is a shortage of dopamine in the brain, resulting from the progressive death of neurons in the *substantia nigra*. In addition, the accumulation of pathological protein deposits, known as Lewy bodies, can further contribute to the development of cognitive symptoms [11].

Cognitive Deficits and Functional Impact

In addition to these disease-specific aspects, it is important to consider the resulting cognitive deficits and their functional impact. Cognitive impairments in neurodegenerative diseases affect core functions such as memory, attention, language, visual processing, and executive abilities [7]. These deficits have a direct impact on everyday activities, for example when taking medication independently, managing finances, maintaining social contacts, or navigating in space [12].

In addition to these practical limitations, emotional and psychosocial burdens are common. Patients often experience frustration, depressive moods, and social isolation, which increase the strain on both those affected and their families.

Therefore, the early detection of cognitive symptoms is of great importance. It provides the basis for targeted interventions and rehabilitation measures that aim to preserve independence and quality of life for as long as possible [13].

2.1.2 Cognitive Rehabilitation: Concepts, Methods, and Target Groups

Multidisciplinary approaches are an important part of cognitive rehabilitation [3]. They combine cognitive training, motor therapy, music therapy, and sensory interventions. The goal is to maintain the independence of patients for as long as possible and to reduce the psychosocial burden on both patients and caregivers.

In addition to traditional training methods, EEG-based biomarkers are increasingly used. Indicators such as the Brain Symmetry Index (BSI) and the Laterality Coefficient (LC) allow an objective assessment of the brain's functional state. EEG analysis makes it possible to adapt rehabilitation strategies to the individual by detecting changes in brain activity, especially in connection with *motor imagery*, which is an established rehabilitation method. These approaches are mainly used with stroke patients, who often suffer from both motor and cognitive impairments.

2. Theoretical Background

Another important method within cognitive rehabilitation is sensory stimulation, with a growing focus on low-frequency vibration. Campbell et al. [14] report in their scoping review that vibrotactile interventions can improve attention, mood, and cognitive stability. Depending on the application context, ranging from short-term interventions in nursing homes to long-term home-based programs, studies found positive outcomes in standardized cognitive tests as well as good patient acceptance.

In summary, cognitive rehabilitation is not limited to one single method but is based on a complementary, multimodal approach. The integration of cognitive training, EEG-based biomarkers, and sensory methods such as vibrotactile stimulation forms a broad field of interventions that is likely to become more important in the future.

Table 2.1 Comparison of studies on tactile low-frequency vibration in dementia management (based on Campbell et al., 2022)

Study	Type of Vibration	Duration / Frequency	Outcomes	Application Context
[15]	Vibroacoustic (40 Hz, music, physioacoustic chair)	2x week, 6 weeks	Improved SLUMS scores, increased attention	Outpatient facility
[16]	Vibroacoustic (40 Hz, daily home application)	Daily, 3 years	Stable MMSE scores over 3 years, reduced frustration	Home use
[17]	Mechanical (WBV, frequency increase from 20–35 Hz)	5x/week, 8 weeks	Significant EEG activation, cognitive improvement	Community centers
[18]	Mechanical (WBV, 30 Hz, 2 mm amplitude)	2x/week, 9 weeks	Improved mobility, balance, high participation rate	Day-care center
[19]	Mechanical (WBV, 30 Hz, 1–2 mm amplitude)	Multiple times/week, 12 min	Good acceptance, some reported side effects (nausea)	Nursing home

2.2 Technical Background

2.2.1 Vibrotactile Stimulation: Principles and Therapeutic Applications

[14]

[15, 19, 18, 20, 17, 16, 21, 22, 23]

2.2.2 Actuation Technologies for Haptic Feedbacks

40 Hz & Gamma Frequenzen, [\[24\]](#) [\[4\]](#) [\[5\]](#) [\[6\]](#) zeigen neurobiologische Wirkung

2.2.3 Voice Coil Actuators for Vibrotactile Stimulation

EEG & Wearables [\[25\]](#) [\[26\]](#) [\[27\]](#) [\[28\]](#) über EEG-Tech, BCI, und mobile Erfassung

2.2.4 Overview of Existing Vibrotactile Stimulation Systems

3 Analysis of the Current VCA-Based System

3.1 Overview of the Current VCA System

This section provides an overview of the existing Voice Coil Actuator (VCA)-based setup. The System consists of seven main parts.

- Spring frame (Minimizing the loss of vertical motion transmitted to the node)
- Magnet Housing (fixed Magnetic field is always formed)
- Bobbin Coil (Magnetic field is formed only when current flows)
- Node (Transmitting vertical motion directly to the human body as sound and vibration)
- Node screw (fixes the node to the bobbin coil)
- Rubber frame (Suppresses vibration from the body from being transmitted to the outside world)
- Connection PCB (Take the analog signal from the AMP and apply it to the bobbin coil)

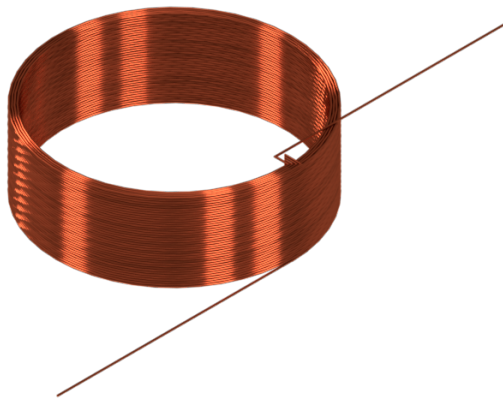


Figure 3.1 coil model

3. Analysis of the Current VCA-Based System

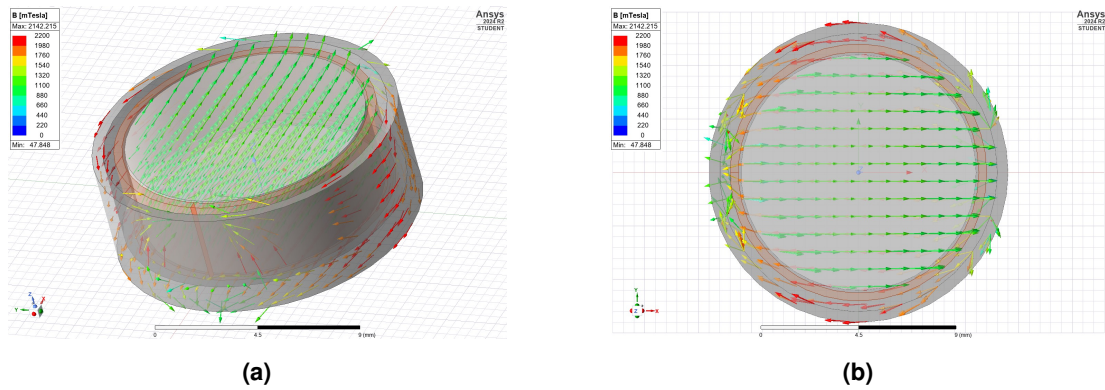


Figure 3.2 Ansys simulation of the B-field in the Magnet Housing
(a) VCA in iso view (b) VCA in top view

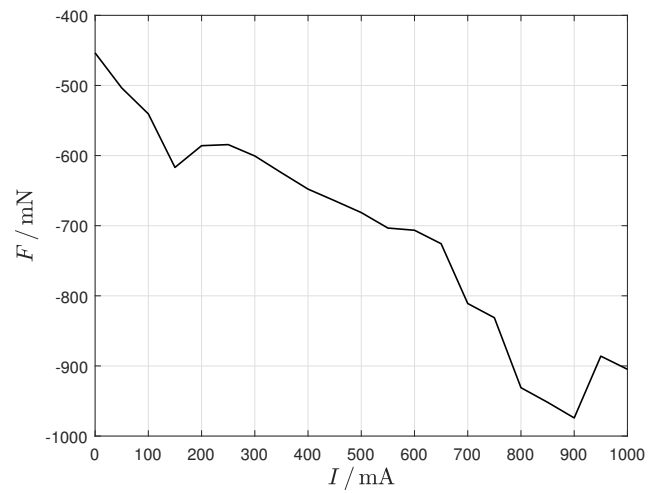


Figure 3.3 Force Plot

3.2 Dynamic Behavior: Frequency Measurement

3.2.1 Objective

3.2.2 Measurement Setup

3.2.3 Results & Interpretation

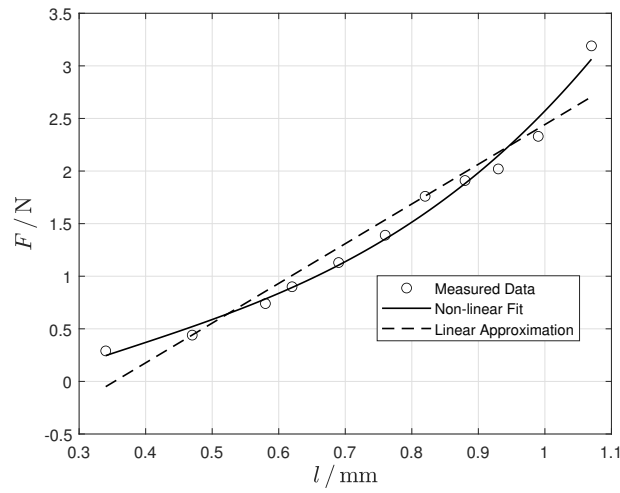


Figure 3.4 SpringTest

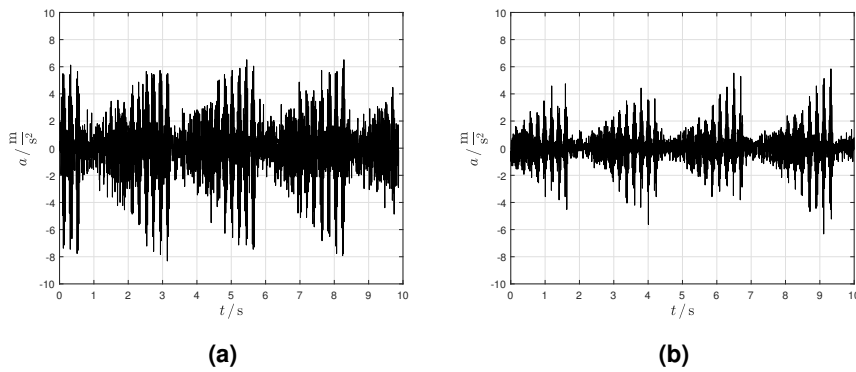


Figure 3.5 (a) With 36g Mass (b) Without Mass

3. Analysis of the Current VCA-Based System

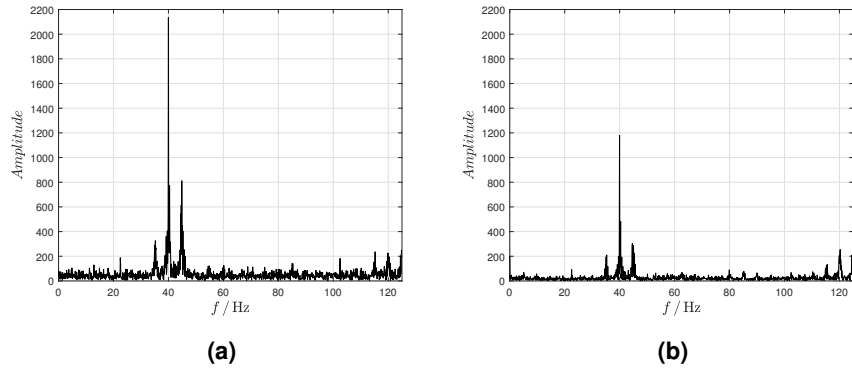


Figure 3.6 (a) With 36g Mass (b) Without Mass

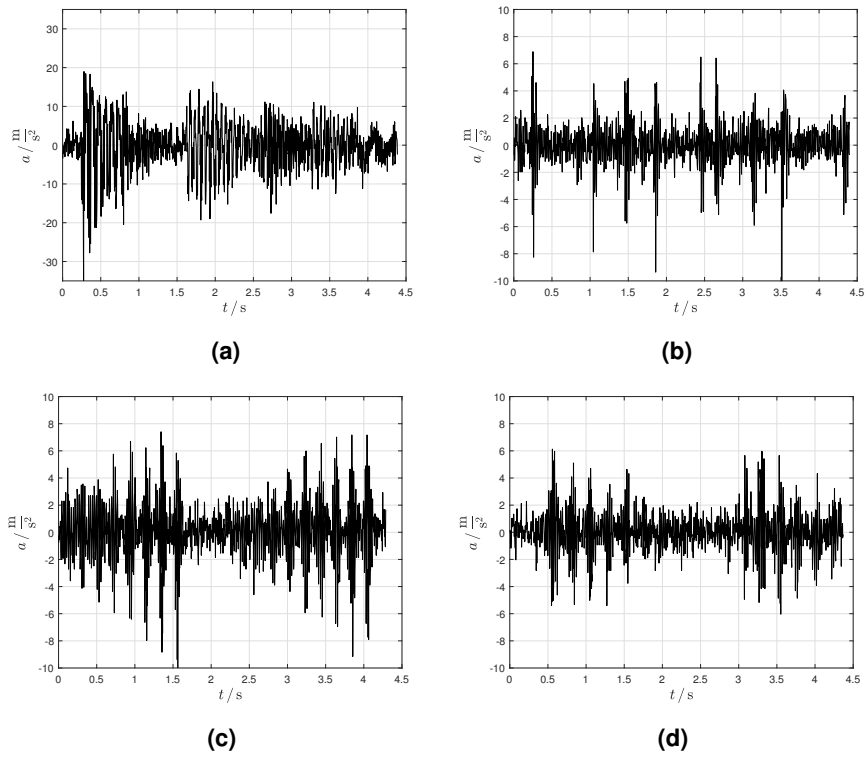


Figure 3.7 (a) Classic Mode (b) Energy Mode (c) Healing Mode (d) Recovery Mode

3. Analysis of the Current VCA-Based System

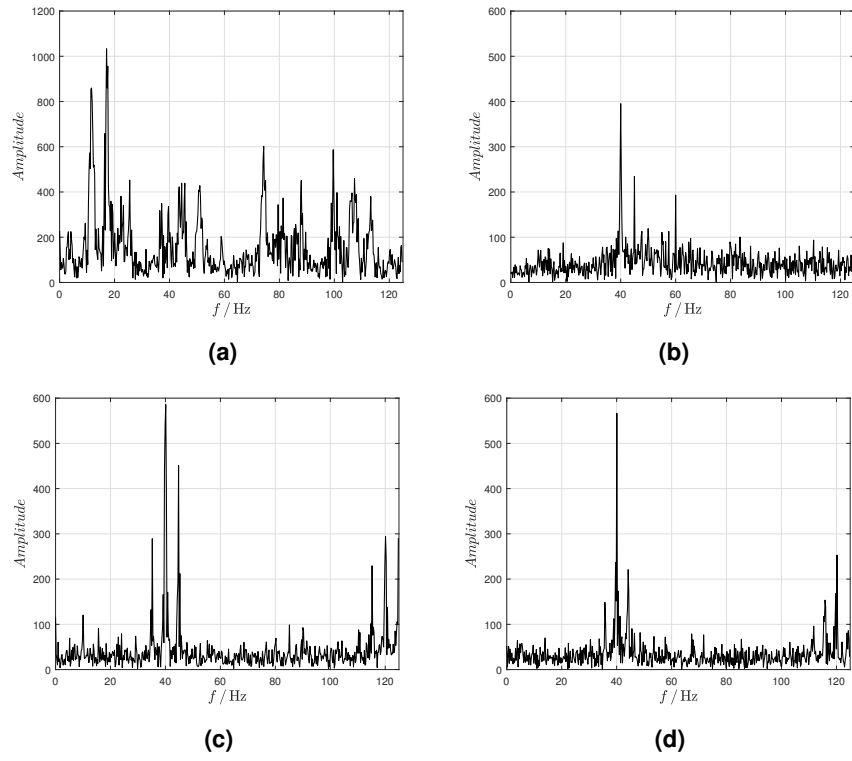


Figure 3.8 (a) Classic Mode (b) Energy Mode (c) Healing Mode (d) Recovery Mode

3.3 Limitations and Identified Challenges

4 Modify

4.1 Designanpassung

4.2 Prototyping

5 Evaluation

6 Conclusion

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